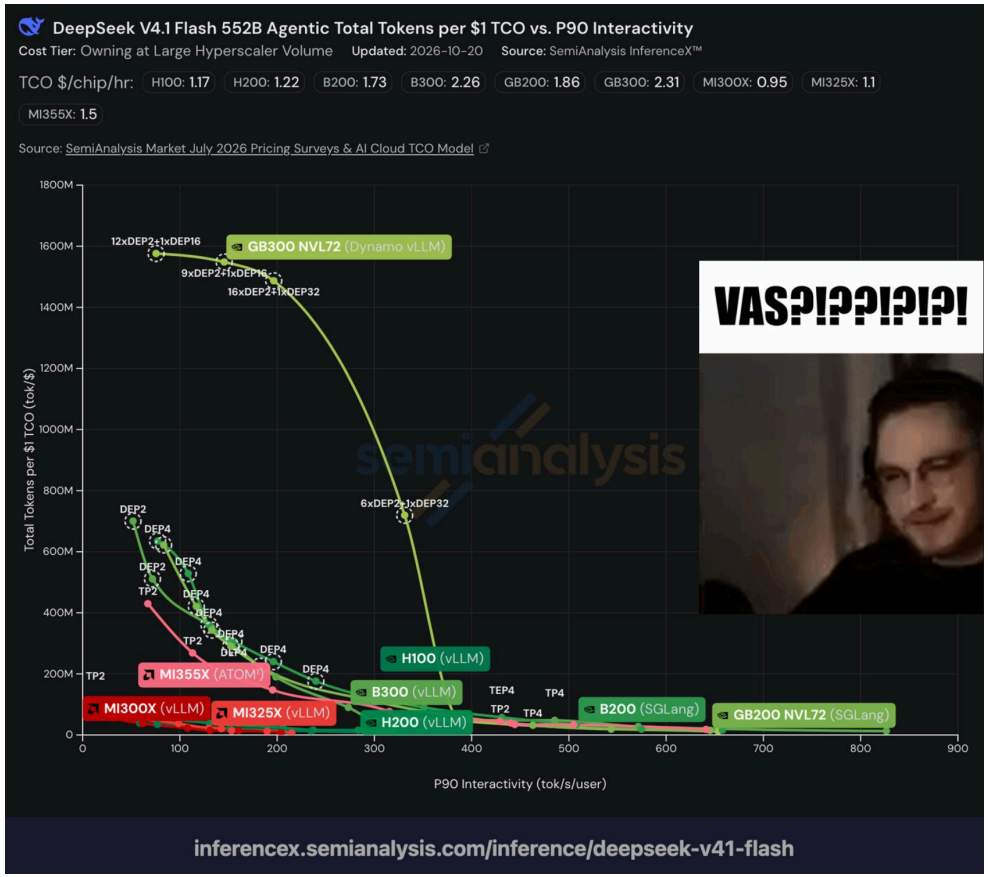
Inference is cool and interesting

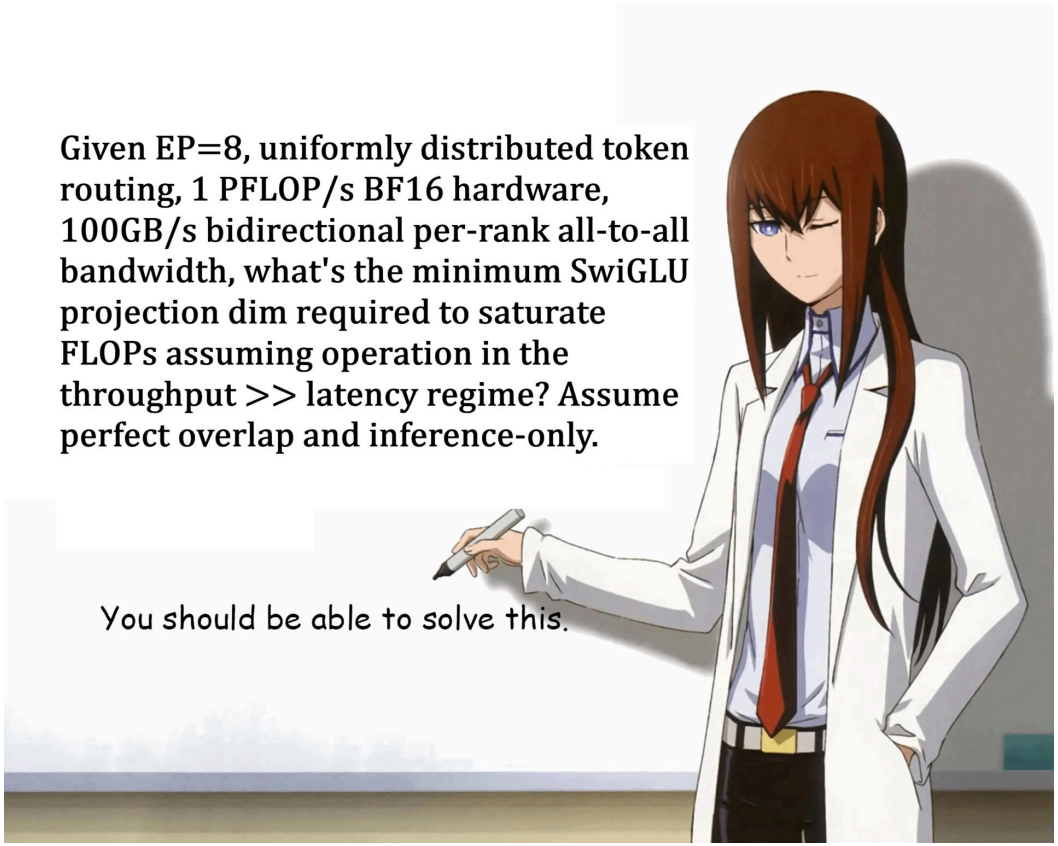
Kart(avya) Vashishtha

October 23, 2026

West Lafayette, IN



big changes happen by the week
many open frontiers



we will have the answer by the end

A forward pass

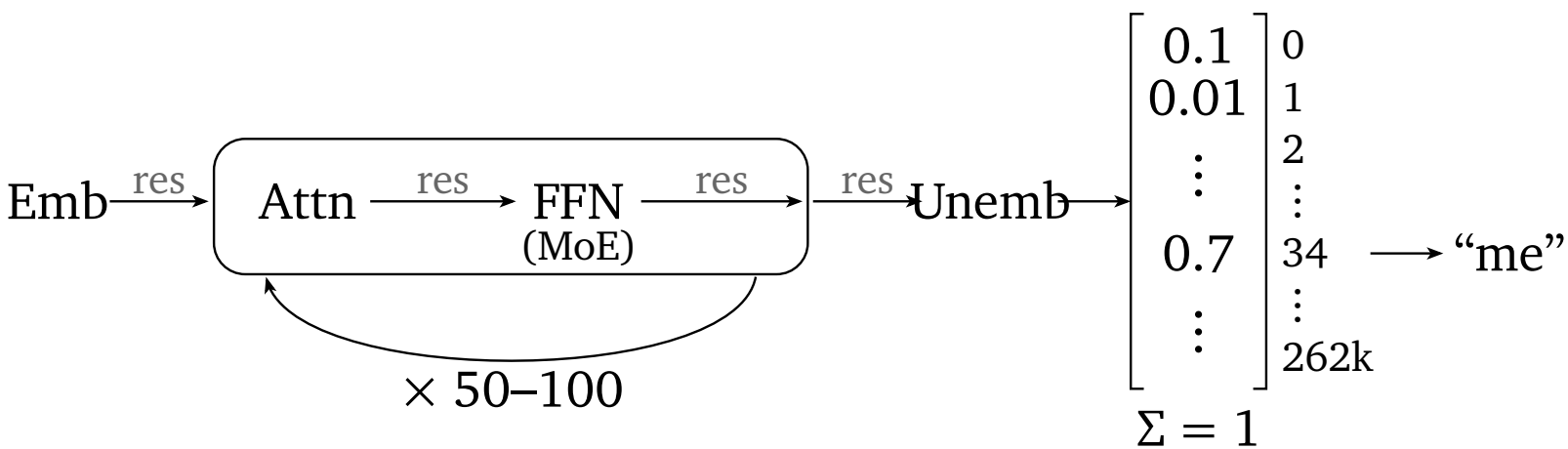
- Embedding
- 100×:
  - Attention
  - FFN
- Unembedding

Embedding

$$[\text{“I”}, \text{“love”}, \text{“bananas!”}] \rightarrow [1, 5, 13] \rightarrow \left[\underbrace{\begin{pmatrix} 3.5 \\ -1.0 \\ \vdots \end{pmatrix}}_{\text{res}}, \begin{pmatrix} \\ \vdots \\ \\ \end{pmatrix}, \begin{pmatrix} \\ \vdots \\ \\ \end{pmatrix} \right]$$

res : [8192], bf16, 2 B per element

Forward pass

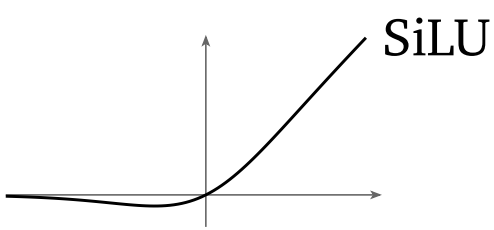


(not paying attention to) Attention

Not relevant in our particular question.

FFN

- $\text{FFN}_{\text{SwiGLU}}(x) = (\text{SiLU}(xW) \odot xV)W_2$
  - Three matmuls, one activation, one elementwise product
- Colloquially: up, gate, down matmuls
- Only matmuls count towards FLOPs



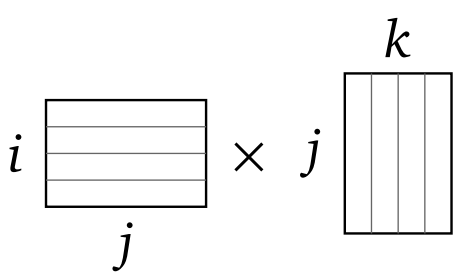
res : [residual_dim], denoted by $[r]$

$$[r] \xrightarrow[\substack{\text{“up”}\\ \text{“gate”}}]{[r;?]} [?] \xrightarrow[\substack{\text{nonlinearity}\\ \text{(elementwise)}}]{f} [?] \xrightarrow[\text{“down”}]{[?;r]} [r]$$

$$\text{e.g. ReLU: } f(x) = \begin{cases} x \text{ if } x \geq 0 \\ 0 \text{ else} \end{cases}$$

(An arrow labeled with a matrix means a matmul with that matrix.)

Question states 1 PFLOP/s. How many FLOPs in a matmul?



$\boxed{2jik}$

	1 token	x tokens
"up" and "gate"	$[1; r] \times [r; ?]$	$[x; r] \times [r; ?]$
"down"	$[1; ?] \times [?; r]$	$[x; ?] \times [?; r]$
FLOPs	$2r? + 2r? + 2?r$	$2r?x + 2r?x + 2?rx$
	$= 6r?$	$= \boxed{6r?x}$

MoE

1. Router: decide which experts
2. Expert matmuls
3. Sum: accumulate expert output

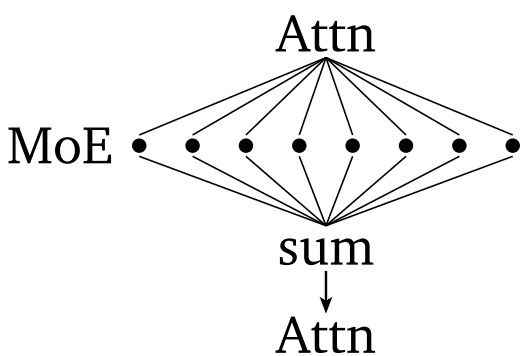
Assume **8** FFN matrices (EP = 8) and 1 expert/token, then the router matmul looks like

$$\text{res} \longrightarrow [r; 8] \longrightarrow \begin{matrix}\begin{bmatrix} 0.1 \\ 0.3 \\ \textcircled{0.5} \\ \vdots \end{bmatrix}\begin{matrix}0\\1\\2\end{matrix} \\ \Sigma = 1\end{matrix} \quad \text{pick FFN } \#2$$

(The projection arrow means a matmul: $[1; \text{res}] \times [r; 8]$.)

FYI: irrelevant here, but important in the real world

general multi-expert-per-token MoE shape



Distributing FFN/MoE onto multiple chips

- Let each chip hold x tokens.
-
-

Distributing FFN/MoE onto multiple chips

- Let each chip hold x tokens.
- Each chip computes its own attention independently.
-

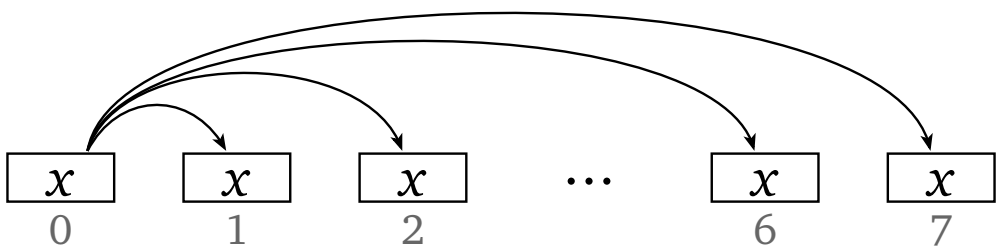
Distributing FFN/MoE onto multiple chips

- Let each chip hold x tokens.
- Each chip computes its own attention independently.
- FFN (MoE): parallelization win!!!!

EP

Calculate communication assuming even token distribution.

8 EP

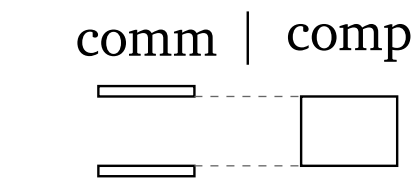


x tokens on a chip: each chip gets $\frac{1}{8}x$ tokens

dispatch outbound data: $7 \times \frac{1}{8}x$

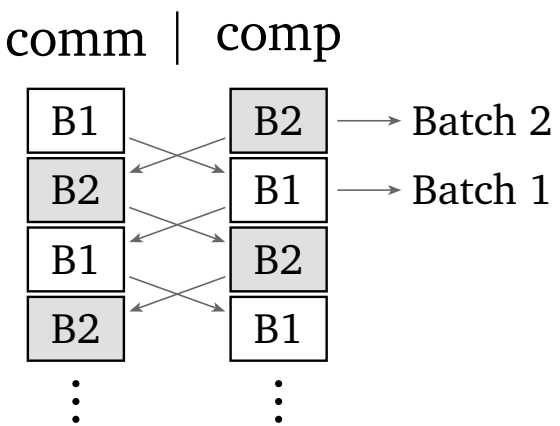
combine outbound: $7 \times \frac{1}{8}x$

Throughput ≫ latency; perfect overlap



Batch holds x tokens

$0.3p + p$ = latency ↓



comm time = computation

$p + p \Rightarrow$ latency ↑, throughput × 2

Final calculation

$$\text{comm time:} \quad \frac{\left(\frac{7}{8}x \text{ tokens} \cdot r \text{ features/token} \cdot 2 \text{ B/feature}\right) \times 2}{100 \text{ GB/s}}$$

$$\text{comp time:} \quad \frac{6r?x \text{ FLOP}}{1 \text{ PFLOP/s}}$$

($\times$ 2 for dispatch and combine)

$$\frac{\left(\frac{7}{8}x \cancel{\text{tokens}} \cdot r \cancel{\text{features/token}} \cdot 2\cancel{B/\text{feature}}\right) \times 2}{100 \text{ G}\cancel{B}/\text{s}} = \frac{6x?r \cancel{\text{FLOP}}}{1 \text{ P}\cancel{\text{FLOP}}/\text{s}}$$

$$\downarrow \ 1000 \text{ TFLOP/s}$$

$$\searrow \ 10^6 \text{ G}\cancel{\text{FLOP}}/\text{s}$$

$$\frac{\frac{7}{2^{\cancel{8}}} \times 2 \times \cancel{2}}{100\,\cancel{\text{G}/s}} = \frac{6?}{10^6\,\cancel{\text{G}/s}}$$

$$\Rightarrow ? = \frac{7}{2} \times \frac{10^6}{100 \times 6}$$

$$= \frac{7}{2} \times \frac{10^4}{6}$$

$$= \frac{7 \times 10^4}{12}$$

$$= \boxed{5833.3\ldots}$$